2. **Answer:** a. Self interactions help regulate or dampen changes or fluctuations in N. By preventing things from changing as quickly, it should make the system be more stable, but it does depend on the details, so need to do the math to really know.
b. Set the differential equations equal to zero. The equation for N^* follows immediately from the $\mathrm{d}P/\mathrm{d}t$ equation. You can then solve the $\mathrm{d}N/\mathrm{d}t$ equations for P and plug in N^* to get P^*

$$N^* = \frac{Z}{\varepsilon\alpha}$$

$$P^* = \frac{r}{\alpha}\left(1 - \frac{Z}{\varepsilon\alpha K}\right)$$

c. Simply identify off-diagonal terms as across species interactions, so they are the coefficients of what multiplies NP, and the identify diagonal elements as self - interaction terms so whatever coefficients multiply N^2 or P^2. Top row is equation for N and bottom two is equation for P.

$$\tilde{J}^* = \mathrm{A} = \begin{pmatrix} -r/K & -\alpha \\ \varepsilon\alpha & 0 \end{pmatrix}$$

d. Just calculate and plug in fixed points as instructed to obtain

$$J^* = \begin{pmatrix} \left.\partial f / \partial N\right|_{\substack{N=N^* \\ P=P^*}} & \left.\partial f / \partial P\right|_{\substack{N=N^* \\ P=P^*}} \\ \left.\partial g / \partial N\right|_{\substack{N=N^* \\ P=P^*}} & \left.\partial f / \partial P\right|_{\substack{N=N^* \\ P=P^*}} \end{pmatrix} = \begin{pmatrix} r - 2r\left(N^* / K\right) - \alpha P^* & -\alpha N^* \\ \varepsilon\alpha P^* & \varepsilon\alpha N^* - Z \end{pmatrix}$$

$$= \begin{pmatrix} -r\dfrac{Z}{\varepsilon\alpha K} & -\dfrac{Z}{\varepsilon} \\ \varepsilon r\left(1 - \dfrac{Z}{\varepsilon\alpha K}\right) & 0 \end{pmatrix}$$

e. For the interaction matrix, we can use the equations from class to quickly calculate the eigenvalues

$$Tr\mathrm{J} = -\frac{r}{K}; \qquad \det \mathrm{J} = \varepsilon\alpha^2$$

$$\lambda_{\pm} = \frac{Tr\mathrm{J}}{2}\left[1 \pm \sqrt{1 - \frac{4\det \mathrm{J}}{(Tr\mathrm{J})^2}}\right] = -\frac{r}{2K}\left[1 \pm \sqrt{1 - \frac{4\varepsilon\alpha^2 K^2}{r^2}}\right]$$

For the standard Jaobian matrix in part d. you can use the equations to obtain

$$Tr\mathrm{J} = -r\frac{Z}{\varepsilon\alpha K}; \qquad \det \mathrm{J} = Zr\left(1 - \frac{Z}{\varepsilon\alpha K}\right)$$

$$\lambda_{\pm} = \frac{Tr\mathrm{J}}{2}\left[1 \pm \sqrt{1 - \frac{4\det \mathrm{J}}{(Tr\mathrm{J})^2}}\right] = -\frac{r}{2}\frac{Z}{\varepsilon\alpha K}\left[1 \pm \sqrt{1 - 4\frac{\varepsilon\alpha K}{rZ}(\varepsilon\alpha K - Z)}\right]$$

f. For the interaction matrix, A, the eigenvalues have a real negative part if

$$1 > \sqrt{1 - \frac{4\varepsilon\alpha^2 K^2}{r^2}}$$

This is always true, so we are done, and we can see system is more stable with carrying capacity and self regulation. For the standard Jacobian, J*, for the real part to be negative we need

$$1 > \sqrt{1 - 4\frac{\varepsilon\alpha K}{rZ}(\varepsilon\alpha K - Z)}$$

which leads to the further condition

$$\varepsilon\alpha K - Z > 0 \Rightarrow K > \frac{Z}{\varepsilon\alpha} = N^*$$

and this is always true because any prey population size *N*, including the fixed point *N**, must be less than the carrying capacity *K*. Obtaining this is easier from the matrix A because less algebra and fewer steps. In both cases we see adding the carrying capacity makes the system more stable.

1. **Answer:**

a. $\dot{N} = \dot{S} + \dot{I} + \dot{R} = -ZR$ (just add equations together)

b. Fixed points are still I,R=0 or S=0 so no nontrival fixed points, meaning no fixed point where S, I, and R are all nonzero.

c. No because the interaction matrix eigenvalues only give the right result when the equilibrium state is nonzero so that we can divide by it, but here the equilibirum state has at least one 0, so we cannot use this approach.

d. Because the population size is changing, the equation for $\dot{R}$ is not redundant, so we must compute a 3x3 matrix and substitute in that $N^*=\gamma/\beta$. Since we are doing the stability analysis now with respect to this fixed point, it would mean that a first infection has already spread through the popultation and pushed the system to this new fixed point before dying out. After this, there would then be a second infection that occurs, and that would be what we are doing the stability analysis for, and the question is whether that second infection would quickly die out or would instead push the system to new fixed point such as S*=0.

$$J^* = \begin{pmatrix} \partial f/\partial S\big|_{\substack{S=N\\I=0\\R=0}} & \partial f/\partial I\big|_{\substack{S=N\\I=0\\R=0}} & \partial f/\partial R\big|_{\substack{S=N\\I=0\\R=0}} \\ \partial g/\partial S\big|_{\substack{S=N\\I=0\\R=0}} & \partial g/\partial I\big|_{\substack{S=N\\I=0\\R=0}} & \partial g/\partial R\big|_{\substack{S=N\\I=0\\R=0}} \\ \partial h/\partial S\big|_{\substack{S=N\\I=0\\R=0}} & \partial h/\partial I\big|_{\substack{S=N\\I=0\\R=0}} & \partial h/\partial R\big|_{\substack{S=N\\I=0\\R=0}} \end{pmatrix} = \begin{pmatrix} -\beta I\big|_{\substack{S=N\\I=0\\R=0}} & -\beta S\big|_{\substack{S=N\\I=0\\R=0}} & 0\big|_{\substack{S=N\\I=0\\R=0}} \\ \beta I\big|_{\substack{S=N\\I=0\\R=0}} & \beta S\big|_{\substack{S=N\\I=0\\R=0}} & 0\big|_{\substack{S=N\\I=0\\R=0}} \\ 0\big|_{\substack{S=N\\I=0\\R=0}} & \gamma\big|_{\substack{S=N\\I=0\\R=0}} & -Z\big|_{\substack{S=N\\I=0\\R=0}} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -\beta N & 0 \\ 0 & \beta N - \gamma & 0 \\ 0 & \gamma & -Z \end{pmatrix} = \begin{pmatrix} 0 & -\gamma & 0 \\ 0 & 0 & 0 \\ 0 & \gamma & -Z \end{pmatrix}$$

e. Eigenvalues are 0, 0, and $-Z$ so eigenvalues are all negative (stable) or 0. Depending on rate constants, system will most likely have some susceptibles become infected and then recover and then all individuals in the recovered compartment eventually die off out of that class. Either the infection will die out and no more will be infected or the infection will eventually push everyone into the recovered class where they will die. Once the infection dies out, I=0, so the last equation becomes $\dot{R} = -ZR \Rightarrow R(t) \propto e^{-Zt}$, and this will go to zero with time, t, so all the recovered individuals will die out.